Balanced Six-Vertex Normalization: Finite-Interval Boundary Certificates and Physical-Norm Error Budgets

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Abstract

We study the normalization mechanism in the specified balanced six-vertex manuscript, retaining unit height jumps, the whole-plane Green kernel $-(2\pi)^{-1}\log|x-y|$, and the ordered balanced-torus limits. For an even real convolution kernel of absolute mass less than one, we prove an explicit exponentially small finite-interval error for the endpoint dressed constant. The proof combines the manuscript's moving-boundary identity with an absolute Neumann-path estimate and also fixes the sign of the half-line limit for signed kernels. For the negative six-vertex kernel this yields one-sided errors, an explicit parameter-dependent majorant, and a rationally checkable truncation budget. A continuous residual and quadrature certificate controls the required endpoint trace; a separate physical-metric certificate encloses finite-volume Kohn stiffness. Neither a finite inverse nor the dressed constant proves the nonlinear Gaussian limit, the sparse-edge inverse bounds, or the physical comparison identifying the height amplitude. Those dependencies and the restricted public literature status are stated explicitly. Contribution assessment: C2/5, useful explicit source-specific error certificates are proved, but the increment is a quantitative extraction of existing contraction and normalization methods rather than a distinct significant new mechanism.

1 Model, metric, and scope

The cached manuscript [1] considers square-lattice six-vertex weights $a = b = 1$ and $0 < c \le 2$. In its balanced torus of dimensions M, L, with L even, each column of horizontal edges has equally many left- and right-pointing arrows. The plane law is the local limit with $M \to \infty$ first and then even $L \to \infty$. This specifies a state; it is not a claim here about every Gibbs state or every slope. The height on the left of an oriented edge exceeds the height on its right by one. Its additive constant is immaterial. Faces are extended using the source's floor convention, and $h_\delta(x) = h(x/\delta)$ for arbitrary real $\delta > 0$.

Set

$$\Delta = 1 - \frac{c^2}{2}, \qquad \lambda = \arccos(-\Delta), \qquad \eta = \frac{\lambda}{\pi}. \tag{1}$$

For $0 < c < 2$, $0 < \lambda < \pi$ and $c = 2\cos(\lambda/2)$. The field normalization in the source is

$$G(x,y) = -\frac{1}{2\pi}\log|x-y|, \quad K_\kappa(x,y) = -2\kappa\log|x-y|, \quad \sigma^2 = 4\pi\kappa = \frac{2}{\pi-\lambda} = \frac{1}{\arcsin(c/2)}. \tag{2}$$

The last equalities are the source's claimed identification, not an independent conclusion of finite calculations in this report. The standard Dirichlet pairing is $\int \nabla u \cdot \nabla v$ without an extra

2π factor. Thus, for smooth compactly supported neutral ϕ, the solution $u = G * \phi$ satisfies $-\Delta u = \phi$ and $\int \phi u = \int |\nabla u|^2$. In particular the factor in (2) is a *variance* multiplier; the field itself is multiplied by σ, not by σ^2.

The whole-plane GFF is defined modulo constants. For measures the source requires compactly supported finite signed neutral μ with

$$\iint \log_+ \frac{1}{|x-y|}\, |\mu|(\mathrm{d}x)|\mu|(\mathrm{d}y) < \infty. \tag{3}$$

This is an ordinary finite integral, not a subtraction of divergent diagonal terms. It excludes atoms. A common smooth mass-one density ρ fixes the representative $h - \langle h, \rho\rangle$; changing the pin changes only the constant, and neutral tests do not see it. Point values of the limiting GFF are not substituted for finite-energy tests.

Three logical levels. First, the source asserts the plane construction, chiral Wick limits, physical normalization, and negative-regularity convergence. Second, this report proves statements about specified continuum integral equations and finite physical operators. Third, applying those statements to a mesh approximation requires stated residual, quadrature, gap, or comparison bounds. We do not collapse these levels. In particular Section 3 does not independently verify the source's Bethe-root estimates or all of its integrable algebra.

2 The dressed equation in the source

The bare kernels are

$$s_\lambda(u) = \frac{\sin\lambda}{\pi(\cosh 2u - \cos\lambda)}, \qquad K_\lambda(u) = \frac{\sin 2\lambda}{\pi(\cosh 2u - \cos 2\lambda)}. \tag{4}$$

At $\lambda = \pi/2$ the latter is identically zero. For a fixed $Q > 0$ write $(K_Q f)(x) = \int_{-Q}^{Q} K_\lambda(x-y) f(y)\,\mathrm{d}y$ and

$$H_Q = (I + K_Q)^{-1}\mathbf{1}, \qquad r_Q = (I + K_Q)^{-1} s_\lambda, \qquad \epsilon_Q = (I + K_Q)^{-1} e', \quad e = -2\pi\eta s_\lambda. \tag{5}$$

The source establishes

$$\int_{\mathbb{R}} K_\lambda = 1 - 2\eta, \qquad h := \int_{\mathbb{R}} |K_\lambda| = |1 - 2\eta| < 1. \tag{6}$$

For completeness, putting $t = e^{2u}$ turns the integral of $(\cosh 2u - \cos\theta)^{-1}$ into $\int_0^\infty [t^2 - 2t\cos\theta + 1]^{-1}\,\mathrm{d}t$. Evaluation by an arctangent, with the sign of $\sin\theta$ retained when $\theta = 2\lambda$ passes π, gives (6). The kernel has constant sign for fixed λ. Hence $I + K_Q$ is invertible on continuous functions in the sup norm by a Neumann series, with inverse norm at most $(1-h)^{-1}$.

In the normalization module the range is $0 < c < 1$, equivalently $2\pi/3 < \lambda < \pi$, so $K_\lambda < 0$. The finite-density, homogeneous-chain calculation gives

$$S_Q := \lim_{L\to\infty} L e_L''(0) = \frac{2}{\pi^2}\frac{\epsilon_Q(Q)}{r_Q(Q)}\, H_Q(Q)^2. \tag{7}$$

Here the filling associated with Q is held fixed before $L \to \infty$; then $Q \to \infty$. The source proves qualitative limits

$$\frac{\epsilon_Q(Q)}{r_Q(Q)} \longrightarrow 2\pi, \qquad H_Q(Q)^2 \longrightarrow b^2 := \frac{\pi}{2(\pi - \lambda)}. \tag{8}$$

The second limit is not an approximation of a bulk value: the bulk constant is $(1 + \int K_\lambda)^{-1} = b^2$, whereas the boundary constant is b. Confusing these two would square the normalization incorrectly.

The source further compares finite stiffness and staggered normalized overlaps with its angular current measure. Its lower comparison uses a filtered inverse of the physical endpoint Hamiltonian, real-time locality, and a sum rule. Its upper comparison uses a two-oriented overlap determinant and widely separated spectral windows with a summable two-window error. Both comparisons are needed to identify κ. Merely evaluating (5) does not supply them.

3 An explicit finite-boundary theorem

This section is independent of the spin model. Assume an even real kernel $K \in C^1(\mathbb{R})$ with $K, K' \in L^1(\mathbb{R})$, and let

$$h = \int |K| < 1, \qquad q_a = \int e^{a|u|}|K(u)|\,\mathrm{d}u < 1 \quad \text{for some } a > 0, \qquad k_0 = \int K. \tag{9}$$

These hypotheses hold for each fixed source parameter $0 < \lambda < \pi$ with some $a > 0$. All constants below can deteriorate as a parameter approaches an endpoint.

Theorem 3.1 (Endpoint error with a signed kernel)**.** *For $Q > 0$, let $H_Q = (I + K_Q)^{-1}\mathbf{1}$ in $C([-Q, Q])$. Then*

$$|H_Q(Q) - b| \le d_Q := \frac{q_a}{1 - q_a} e^{-2aQ}, \qquad b = (1 + k_0)^{-1/2} > 0. \tag{10}$$

If $K \le 0$, then $H_Q(Q)$ increases with Q and $1 \le H_Q(Q) \le b$; consequently $0 \le b - H_Q(Q) \le d_Q$.

Proof. Reflection and translation $x = Q - t$ move the interval to $[0, 2Q]$, and evenness changes $K(y - x)$ back to $K(x - y)$. Let T_R be convolution restricted to $[0, R]$, and T_+ its half-line counterpart on $C_b([0, \infty))$. The norm of each is at most h. The endpoint is

$$H_Q(Q) = \sum_{n=0}^{\infty} (-1)^n (T_{2Q}^n \mathbf{1})(0).$$

For $n \ge 1$, write the half-line term as an integral over $x_1, \dots, x_n \ge 0$ with $x_0 = 0$ and integrand $\prod_{j=1}^n K(x_{j-1} - x_j)$. Its difference from the finite-interval term is supported on paths having at least one $x_j > 2Q$. Such a path satisfies $\sum_j |x_j - x_{j-1}| > 2Q$. Therefore its absolute contribution is at most

$$e^{-2aQ} \int_{\mathbb{R}^n} \prod_{j=1}^{n} e^{a|x_j - x_{j-1}|} |K(x_j - x_{j-1})|\,\mathrm{d}x_1 \cdots \mathrm{d}x_n = e^{-2aQ} q_a^n.$$

The enlargement to $\mathbb{R}^n$ is applied only after taking absolute values; the triangular increment change of variables has determinant one. Summing gives

$$|H_Q(Q) - B| \le d_Q, \qquad B = ((I + T_+)^{-1}\mathbf{1})(0). \tag{11}$$

We identify B, including its sign. Define $F(Q) = \int_{-Q}^{Q} H_Q(x)\,\mathrm{d}x$. The inverse depends differentiably on the two moving endpoints on compact Q intervals; this also follows by rescaling to $[-1, 1]$ and differentiating the uniformly convergent inverse series. Differentiating the equation, then using symmetry of its kernel, gives

$$\begin{aligned} \partial_Q H_Q &= -(I + K_Q)^{-1}\{K(\cdot - Q)H_Q(Q) + K(\cdot + Q)H_Q(-Q)\}, \\ F'(Q) &= 2H_Q(Q) - 2H_Q(Q) \int_{-Q}^{Q} H_Q(x)K(x - Q)\,\mathrm{d}x \\ &= 2H_Q(Q)^2. \end{aligned} \tag{*}$$

Indeed H_Q is even, the inverse is self-adjoint on $L^2(-Q,Q)$, and the integral on the second line is $1 - H_Q(Q)$ by evaluating the original equation at Q. The L^2 and continuous inverses agree on this continuous right-hand side.

For each fixed n, the average over $x_0 \in [-Q,Q]$ of the n-fold restricted convolution tends to k_0^n. To check this without a boundary assumption, introduce the increments $z_j = x_j - x_{j-1}$. For fixed increments, the fraction of initial points for which all intermediate points remain in $[-Q,Q]$ tends to one and lies in $[0,1]$. Dominated convergence with $\prod_j |K(z_j)|$ proves the assertion. Since the averaged absolute nth term is at most h^n, summing the series yields

$$\lim_{Q\to\infty} \frac{F(Q)}{2Q} = \sum_{n=0}^{\infty} (-k_0)^n = \frac{1}{1+k_0}.$$

But (*) and (11), together with $F(0) = 0$, give the same limit as B^2. Thus $B^2 = (1+k_0)^{-1}$. Apply this reasoning to tK for $0 \le t \le 1$. Its half-line endpoint depends continuously on t by the uniform Neumann series, never vanishes because its square is $(1+tk_0)^{-1}$, and equals one at $t = 0$. This proves the positive sign. When $K \le 0$, every term is nonnegative and its domain increases with Q. This proves monotonicity and the remaining statements. □

What is incremental here. The shape identity (*) and the qualitative half-line identification are already in the source normalization proof. Neumann inversion and exponential path bounds are classical. The local addition is the displayed, directly usable error constant $q_a/(1-q_a)$, its one-sided specialization, and its combination below with continuous residual and quadrature errors. We do not claim historical novelty for the half-line identity or for contraction estimates.

Corollary 3.2 (A uniform parameter set)**.** *If a family of kernels satisfies the hypotheses with common $a > 0$, $q_a \le q_* < 1$, and $h \le h_* < 1$, then (10) is uniform with $q_*/(1-q_*)$. For the source this is available on compact subintervals of $0 < \lambda < \pi$, but the displayed bounds give no uniform control as $c \downarrow 0$ or $c \uparrow 2$.*

Proof. Only these bounds were used in the estimates. The source kernel is smooth, exponentially decaying uniformly on compact parameter sets; continuity of its weighted mass at $a = 0$ gives a common sufficiently small positive a. At either limiting parameter the contraction margin $1 - |1 - 2\lambda/\pi|$ vanishes. □

4 Certified point traces, not just matrix residuals

Theorem 4.1 (Continuous and sampled residual budgets)**.** *Under Section 3, let $u \in C([-Q,Q])$ and suppose $\|(I+K_Q)u - \mathbf{1}\|_\infty \le \varepsilon$. Then*

$$|u(Q) - b| \le E_Q := d_Q + \frac{\varepsilon}{1-h}. \tag{12}$$

For a real trial define $u_- = \max(0, u(Q) - E_Q)$ and $u_+ = \max(0, u(Q) + E_Q)$. The exact algebraic normalization $4b^2/\pi$ belongs to $[4u_-^2/\pi, 4u_+^2/\pi]$.

Here is a sufficient fully specified way to certify ε. Assume $u \in C^1$, $\|u\|_\infty \le U$, $\|u'\|_\infty \le U_1$, and put $k_1 = \|K'\|_1$. Partition $[-Q,Q]$ into cells of maximum length τ, with midpoint y_j and length w_j. Use test nodes x_i of covering radius ρ on the closed *interval. Suppose certified arithmetic bounds the absolute residual at all these nodes, computed with midpoint quadrature, by ζ. Then it suffices to take*

$$\varepsilon = \zeta + \frac{\tau}{2}(k_1 U + h U_1) + \rho(U_1 + k_1 U). \tag{13}$$

Arithmetic error must already be included in ζ.

Proof. Subtract the exact equation and apply the inverse norm $(1-h)^{-1}$. Then use Theorem 3.1. Squaring the resulting positive interval for b proves the stated enclosure. For the quadrature formula fix x and let $f(y) = K(x-y)u(y)$. On a cell I with midpoint y_I,

$$\left| \int_I (f(y) - f(y_I))\,\mathrm{d}y \right| \le \frac{|I|}{2} \int_I |f'(y)|\,\mathrm{d}y.$$

This follows by integrating the fundamental theorem of calculus on each side of the midpoint; each derivative is counted over a segment of length at most $|I|/2$. Summing cells bounds the integration error by $\tau(k_1 U + h U_1)/2$, uniformly in x. The exact continuous residual has derivative bounded by $U_1 + k_1 U$. Move any x to a test node within distance ρ to obtain (13). No sample-only bound is used. □

Remark 4.2 (The required function space). The source's root evaluation operators act on actual continuous representatives, or on C_b^1, not on equivalence classes in Lebesgue L^∞. Theorem 4.1 likewise uses a pointwise norm. An L^2 residual alone cannot control this endpoint: take a continuous triangular spike v_ϵ of height one at Q, supported on $[Q-\epsilon, Q]$. Then $\|v_\epsilon\|_2^2 = \epsilon/3$ and Young's inequality gives $\|(I+K_Q)v_\epsilon\|_2 \le (1+h)\sqrt{\epsilon/3}$, while $v_\epsilon(Q) = 1$. Adding this spike to H_Q proves the failure of an L^2-only endpoint certificate. Derivative or Sobolev trace bounds can repair that failure, but are additional hypotheses.

Corollary 4.3 (Integral Neumann truncation)**.** *Let* $u_N = \sum_{j=0}^{N} (-K_Q)^j \mathbf{1}$. *Then*

$$\|(I+K_Q)u_N - \mathbf{1}\|_\infty \le h^{N+1}, \qquad \|u_N - H_Q\|_\infty \le \frac{h^{N+1}}{1-h}.$$

For $K \le 0$, $u_N(Q) \le H_Q(Q) \le b$, *so*

$$0 \le b - u_N(Q) \le d_Q + \frac{h^{N+1}}{1-h}. \tag{14}$$

Proof. The residual series telescopes to $(-1)^N K_Q^{N+1}\mathbf{1}$. Use the contraction norm. In the negative-kernel case all terms of the inverse series are nonnegative. □

An integral Neumann truncation is not automatically a finite arithmetic algorithm. Its iterated integrals still need enclosure. If an evaluated continuous trial $\widetilde{u}_N$ has a certified sup error ν_N from u_N, add ν_N to (14); its sign need no longer be one-sided. Similarly, midpoint weights in (13) must not be silently identified with the source's sparse Bethe weights $\ell_j = 1/\xi'(y_j)$.

5 An explicit six-vertex error budget

Write $\mu = \pi - \lambda \in (0, \pi/3)$ in the source's $c < 1$ range. Then

$$k(u) := -K_\lambda(u) = \frac{\sin 2\mu}{\pi(\cosh 2u - \cos 2\mu)}, \qquad h = 1 - \frac{2\mu}{\pi}.$$

Define

$$D = \min(1, 1-\cos 2\mu) > 0, \quad C = \frac{2\sin 2\mu}{\pi D}, \quad \delta = 1 - h.$$

Since $\cosh 2u - \cos 2\mu \ge D \cosh 2u$, $k(u) \le C e^{-2|u|}$. Thus for $0 < a < 2$,

$$q_a \le h + C \int_{\mathbb{R}} (e^{a|u|} - 1) e^{-2|u|}\,\mathrm{d}u = h + \frac{Ca}{2-a}. \tag{15}$$

Choosing $a = \delta/(C+\delta)$ makes the extra term $C\delta/(2C+\delta) < \delta/2$. This provides an explicit admissible exponential weight, rather than an unspecified sufficiently small one. It is conservative near the parameter boundary.

At the concrete parameter

$$\lambda = \frac{3\pi}{4}, \quad c = \sqrt{2-\sqrt{2}}, \quad K_\lambda(u) = -\frac{1}{\pi \cosh 2u}, \quad h = \frac{1}{2}, \quad b = \sqrt{2}, \tag{16}$$

we can choose $a = 1/3$. Using $\pi > 3$ in (15) gives $q_a \le 19/30$, hence

$$d_Q \le \frac{19}{11} e^{-2Q/3}.$$

With $Q = 15$ and $N = 20$,

$$0 \le \sqrt{2} - u_{20}(15) \le \frac{19}{11} e^{-10} + 2^{-20} < \frac{1}{12500}. \tag{17}$$

The strict last inequality has a rational certificate: replace e^{10} in its denominator by $\sum_{j=0}^{40} 10^j/j!$ and cross-multiply positive integers. The owned checks perform exactly that comparison, not a floating point exponential comparison. No numerical value of the unevaluated twentyfold integral is claimed. The corresponding algebraic coefficient is $4b^2/\pi = 8/\pi$. If $0 \le b - v \le e$, then

$$0 \le \frac{4}{\pi}(b^2 - v^2) \le \frac{8b}{\pi} e. \tag{18}$$

At (16), the bound $\sqrt{2} < 3/2$, $\pi > 3$ turns (17) into a coefficient error less than $1/3125$. Again this certifies the algebraic normalization target, not a rate of mesh convergence of the height law.

Proposition 5.1 (Keeping the other stiffness error)**.** *In the negative-kernel range suppose, in addition to the source identity* (7)*, a separate certificate gives* $|R_Q - 2\pi| \le e_R$*, where* $R_Q = \epsilon_Q(Q)/r_Q(Q)$*. Then*

$$\left| S_Q - \frac{2}{\pi - \lambda} \right| \le \frac{2}{\pi^2} \left(b^2 e_R + 4\pi b\, d_Q \right). \tag{19}$$

Proof. Subtract $(2/\pi^2)(2\pi)b^2$ in (7). Use $H_Q(Q)^2 \le b^2$ and $b^2 - H_Q(Q)^2 \le 2b\, d_Q$. □

The report does not supply a numerical e_R for the source's dressed energy-density ratio. Both numerator and denominator decay at the edge, so absolute errors without a denominator lower bound are insufficient. This is a genuine remaining requirement, not an error that can be absorbed into d_Q by notation.

6 Finite physical stiffness and Gram metrics

The source's finite-volume Kohn identity is particularly sensitive to the Hilbert metric. Fix the magnetization sector and let Ω be its normalized simple ground state. On $\Omega^\perp$ let $A = H_L - e_L \ge gI$ with a certified $g > 0$. Write $j = P_{\Omega^\perp} J_L \Omega$ and let D denote the source's diamagnetic expectation. In its seam-twist convention,

$$L e_L''(0) = D - \frac{2}{L} \left\langle j, A^{-1} j \right\rangle. \tag{20}$$

The inverse excludes the ground state and is not an inverse on all magnetization sectors. We use inner products conjugate-linear in their first argument.

Proposition 6.1 (A physical residual enclosure)**.** *For any* $v \in \Omega^\perp$*, set* $r = j - Av$ *and* $T = 2\,\mathrm{Re}\,\langle j, v\rangle - \langle v, Av\rangle$*. Then*

$$\left\langle j, A^{-1} j \right\rangle = T + \left\langle r, A^{-1} r \right\rangle, \qquad D - \frac{2}{L}\left(T + \frac{\|r\|^2}{g} \right) \le L e_L''(0) \le D - \frac{2T}{L}. \tag{21}$$

This is a finite-volume statement; its utility along a sequence requires actual control of g *and of the full residual.*

Proof. Put $w = A^{-1}j - v = A^{-1}r$. Expanding $\langle v + w, A(v + w)\rangle$ and the definition of T gives the exact identity. The inequalities follow from $0 < A^{-1} \le g^{-1}I$. □

Suppose a trial space has injective synthesis $J : \mathbb{C}^m \to \Omega^\perp$ and metric $G = J^*J > 0$. Its orthogonal projector is $P = JG^{-1}J^*$, its stiffness matrix is $K = J^*AJ$, and its coordinate operator is $B = G^{-1}K$, not K itself. For $v = Jz$ and load $\ell = J^*j$,

$$\|j - AJz\|^2 = (\ell - Kz)^*G^{-1}(\ell - Kz) + \|(I - P)(j - AJz)\|^2 . \tag{22}$$

Indeed these are the squared norms of the orthogonal components of the full residual. The first term alone is only a projected certificate. Likewise $GB = B^*G$ expresses physical self-adjointness; ordinary Euclidean coordinate norms do not.

Here is an exact small example on an already reduced space:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}, \quad j = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad J = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad z = \frac{1}{2}.$$

The projected residual is zero, but the full residual is $(0, 1)^t$. Here $T = 1/2$ and $\langle j, A^{-1}j\rangle = 7/10$. With $g = 2$, the certified interval is $[1/2, 1]$, not the false singleton $\{1/2\}$. Rescaling the trial basis to $J' = (2, 0)^t$ gives $G' = 4, K' = 8, \ell' = 2, z' = 1/4$ and leaves every physical quantity unchanged. This illustrates the metric correction and the unresolved complement separately.

The edge inverse is not replaced. In the thermodynamic source the root-weighted Gaudin inverse is uniform on pointwise function spaces only after a bulk/edge argument. The bulk is contracting; sparse edge indices require compactness and a maximum principle for a limiting one-sided averaging equation whose deficit can vanish. Exponential conjugation is then a small operator perturbation. Strict diagonal dominance at every finite size does not give a size-independent inverse bound. For example $A_n = \mathrm{diag}(1, n^{-1})$ is strictly diagonally dominant and invertible, but $\|A_n^{-1}\| = n$. Neither (21) nor a finite Gram calculation proves the source's edge theorem. In particular the positive gap g above is an input, not a uniform spectral gap assertion for the critical chain.

7 The original Ram arguments and their proper use

The private originals were read locally and are not used as authority for a six-vertex theorem. Their role is methodological and specific. The physical-metric chapter [6] derives $G = J^*J$, $P = JG^{-1}J^*$, and the decomposition of the full residual into projected and omitted components. Applied to the source's actual Kohn quadratic form, these give (22) and explain why Euclidean conditioning of a coefficient matrix is not the physical error. A relative metric estimate such as $(1 - \epsilon)G \le \widehat{G} \le (1 + \epsilon)G$ is meaningful; entrywise closeness without a lower frame bound is not.

The original spectral-limits chapter [7] constructs a graph-core approximation with a spurious moving eigenvalue and gives two-sided inverse and dual-target residual criteria. It distinguishes strong resolvent convergence from norm-resolvent convergence. This is exactly the distinction needed before asserting a uniform edge inverse from finite matrix checks. Its dual-target formula also explains the trace issue in Remark 4.2: an evaluation functional is bounded on $C([-Q, Q])$, but is not bounded on L^2. Changing the norm changes the admissible target, not just the size of an error bar.

The original integral-transform report [8], in the passages read here, proves a coordinate localization by $\beta = x - \alpha$, including the moving integration limits and the sign of the Jacobian. Its point-jet passages require differentiable representatives and justified boundary traces. These facts support the exact domain change $[-Q, Q] \to [0, 2Q]$ in our path proof. An inverse coordinate change of kernels is not an inverse of the integral operator; the latter is proved separately by

the contraction hypothesis in (9). The local result therefore uses the Ram discipline of physical reconstruction, domain control, and residual accounting without renaming classical Neumann inversion as a new theory.

8 Source architecture and retained obligations

The following map covers all nine scientific modules. A map is not a claim to have read every line of each module. The subsequent scope table gives the actual literal reads; deeper inputs outside those intervals remain imported.

Module	Scientific role	Obligation retained here
01	Balanced plane law; field and test conventions	Ordered torus limits; ordinary logarithmic energy; pinned and modulo-constant formulations.
02	Physical gauge, finite rows, Bethe vectors, scalar products	The staggered physical gauge and adjoint, marked rows, deleted-column identities, Hilbert duality and normalized overlap are not replaced by an arbitrary matrix model.
03	Root density, edge inverses, harmonic row normalization, fixed-word limits and clustering	Uniformity must include sparse edges. Array limits have $B \to \infty$ at fixed small spacing d, followed by $d \to 0$; word length stays fixed.
04	Physical plane, angular spectrum, low bands, Hamiltonian, sum rule and variogram	The local endpoint generator E differs from the damping generator H_0. The sum rule is initially away from zero momentum; division by its vanishing factor is not valid at zero.
05	Homogeneous stiffness and staggered overlap matching	Fixed filling before half filling; the ratio in (8); infrared filtering; exact two-dip bounds and both orientations of the finite trace calculation.
06	Chiral limits, collision coefficients and Wick recursion	Two-scale limits, holomorphic continuation and collision estimates precede the removal of artificial poles. Covariance alone is not Wick's theorem.
07	Ordinary moments, desmoothing, finite-energy measures and negative topologies	Moment domination and separate tightness arguments are required. Restriction, outward-scaling, and intrinsic dual norms are distinct.
08	Assembly of the claimed theorem	For $c < 1$, the two normalization comparisons; for $1 \le c < 2$, the conditional free-energy amplitude theorem only after GFF convergence.
09	Endpoint $c = 2$, FK rotation and marked correlations	The endpoint uses separate geometric/correlation inputs; it is not a uniform limit of the contractive constants in this report.

Details that must not be discarded. The source's finite seam derivative yields (20), not an arbitrary Hessian in a changed gauge. Its current sum rule is $2\,\mathrm{sym}\, p_*(E\mu) = D_0\ \mathrm{d}p/(2\pi)$ away from $p = 0$. The exact variogram is an orbit integral, with asymptotic $4\kappa \log|x| + O(1)$, whereas the limiting covariance kernel is $-2\kappa \log|x - y|$. These factors agree with (2). The source retains unit height jumps and subtracts the vacuum energy. The mixed current contractions vanish off the diagonal only after the chiral construction. They are not a consequence of solving (5).

For ordinary finite-energy measures the source uses a four-variogram identity; all four copies of the divergent mesh logarithm cancel by neutrality. Distinct continuum points that round to

the same face are retained in the bound. Passing from the resulting covariance limit to Gaussian laws still requires approximation by smooth neutral tests. For negative regularity, the source localizes a common pin and obtains higher moments before using wavelet tails. Its restriction norm on an arbitrary bounded open set uses the quotient definition, not an unavailable boundary extension operator. The outward convention tests $f(x/\varepsilon)$ and reaches increasing boxes; it is not silently replaced by a usual local restriction norm. The intrinsic dual-space and endpoint topology proofs are outside the literal proof scope recorded below.

Corrections and non-corrections. No upstream source formula is corrected by this report. In the normalization chain inspected, the bulk/boundary distinction, the π factors, the seam convention, and the ordered limits are retained. The report prevents several possible *misreadings*: a finite inverse does not prove edge uniformity; a regular midpoint rule is not the sparse Bethe quadrature; an L^2 residual is not an endpoint trace; and a coefficient identity is not a nonlinear universality proof. Absence of a detected defect in this scoped work does not certify the entire manuscript.

9 Current public comparison

The public-primary refresh was made on October 9, 2026, using public titles and URLs only. The March 6 preprint of Duminil-Copin–Kozlowski–Lammers–Manolescu [2] states full-plane GFF convergence for isotropic $c \in [\sqrt{3}, 2]$, equivalently $\Delta \in [-1, -1/2]$, and an anisotropic extension with a suitable embedding. That citation does not establish the entire $0 < c < 2$ range.

The June 30, 2026 version of the rotational-invariance paper [3] concerns critical random-cluster models with $1 \le q \le 4$ and associated Potts correlations. Its abstract describes quantitative and angular-uniformity improvements. The Bethe-root/free-energy paper [4], version 2 of August 24, 2022, concerns existence and condensation of roots and torus free energy for $\Delta < 1$. A free-energy theorem is not by itself a proof of Gaussian height fluctuations.

The October 5 preprint of Averous and collaborators [5] states Dirichlet-GFF convergence for simply connected domains with piecewise constant boundary data in stated ranges including Δ near zero and $[-1, -1/2]$. Its abstract explicitly uses full-plane convergence and reflection positivity. It is a separate domain statement, not a license to turn the whole-plane pin used here into zero Dirichlet data on an arbitrary domain. Only the primary abstracts and version metadata of these four external papers were reviewed for this comparison; their full proofs are not claimed as audited. Our local theorems above are proved here without invoking their global conclusions.

The assigned September 23 manuscript [1] asserts the broader balanced-state theorem. We distinguish that assertion from the external results just described and from the local results of this report. Public availability and captured byte identity do not constitute independent proof verification.

10 Evidence, costs, and further work

10.1 Literal scope

The fresh read ledger records pre-read hashes, displayed bounded intervals, post-read hashes, and unread complements for every input. For public scientific modules the intervals are:

File	Literal lines read
`01-introduction.tex`	1–265
`02-finite-rows.tex`	1–260
`03-thermodynamic-kernels.tex`	1–190, 470–769
`04-angular-plane.tex`	819–1086
`05-normalization.tex`	1–847 (complete)
`06-chiral-gaussianity.tex`	778–903
`07-probabilistic-endgame.tex`	1–290, 556–865
`08-main-proof.tex`	1–55 (complete)
`09-endpoint-rotation.tex`	517–558

The main file (28 lines) and bibliography (368 lines) were also read in full. This totals 3349 unique public-source lines. The README, preamble, and figure code were inventoried but not literally read. In particular no claim is made to have inspected an included figure merely because its reference was seen.

The fresh private proof reads comprise the complete 210-line physical-metric chapter and complete 242-line spectral-limits chapter, and lines 120–315 and 984–1126 of the 1794-line original TR117. That is 791 unique private lines. The remaining TR117 lines, its external included files, and the rest of the private corpus were not read. None was transmitted to a public service. The scope is deliberately narrower than a full audit of the approximately ninety-page scientific source. The whole source architecture is mapped, but the uninspected determinants, root asymptotics, low-band estimates, chiral precursor bounds and endpoint geometry remain imported wherever invoked.

10.2 Reproducible finite checks

The owned script `M425-exact-checks.py` uses rational arithmetic and runs independently of external checkers. Its output distinguishes parameter cases from assertions:

Fixture group	Cases	Assertions
Weighted discrete path escape inequalities	1152	2304
Explicit exponential-weight choices	20	60
Source parameter truncation budget	1	4
Physical Kohn identity and Gram residual	180	720
Midpoint integration error on a compact C^1 kernel	816	1632
Neutral metric and amplitude-squared identities	70	140
Total	2239	4860

The path fixtures are a discrete analogue of the escape estimate; they do *not* assert a discrete square-root endpoint identity. The quadrature fixtures integrate polynomials exactly, using a compactly supported C^1 kernel, and test the constants rather than the actual Bethe-root mesh. The rational exponential check certifies (17) without evaluating $H_{15}(15)$. These tests supplement the proofs; they do not validate the upstream thermodynamic theorem, Gaussianity, spectral gap, or all-parameter universality. Independent reviews are not added to the author's test count.

10.3 Advantages, limitations, and extensions

The boundary theorem supplies a usable accuracy target in Q before choosing any representation of the integral operator. The negative kernel gives monotone lower approximations and avoids unnecessary two-sided uncertainty. A continuous residual can then accommodate different bases or validated quadrature methods. The physical Kohn enclosure is likewise basis-invariant when the Gram metric and omitted residual are retained.

These advantages have explicit costs. The factor $(1-h)^{-1}$ and the admissible weight deteriorate near the endpoints. Derivative enclosures in (13) may be pessimistic, and high-order

quadrature does not remove the obligation to control a point trace. The finite stiffness certificate can be weak when its genuine finite-volume gap is small. No runtime improvement or measured speedup is asserted.

One feasible extension is to optimize a numerically while bounding q_a by interval arithmetic; the proof only needs a certified $q_a < 1$. Another is to replace the midpoint estimate by validated piecewise polynomial integration with an explicit derivative remainder. The more consequential unfinished step is a relative certificate for $\epsilon_Q(Q)/r_Q(Q)$, including a positive denominator bound, followed by quantitative physical comparison errors in the prescribed limit order. Uniform sparse-edge inverses, the staggered overlap trace density, and higher-current limits cannot be supplied by the finite algebra here.

11 Conclusion

The local contribution is a quantitative and reproducible normalization interface: an exponentially accurate dressed endpoint, an honest pointwise residual budget, and a separate physical stiffness enclosure. It preserves the source's height convention and all factors connecting Dirichlet energy, covariance, variogram, and stiffness. Its C2 assessment concerns the modest increment beyond existing contraction and shape-derivative methods, independently of L3 applicability. A global theorem is not required for a higher contribution score, but finite restatements of classical methods are not assigned such credit here. The complete nonlinear GFF claim and its uninspected machinery remain outside the proved local results.

References

[1] OpenAI, *The Gaussian free field limit of the balanced six-vertex model with variance multiplier* $1/\arcsin(c/2)$, September 23, 2026. Assigned cached manuscript, build modules 01–09. Public repository: https://github.com/openai/math.

[2] H. Duminil-Copin, K. K. Kozlowski, P. Lammers, and I. Manolescu, *Gaussian free field convergence of the six-vertex model with* $-1 \leq \Delta \leq -1/2$, March 6, 2026, version 1. https://arxiv.org/abs/2603.06268v1.

[3] H. Duminil-Copin, K. K. Kozlowski, D. Krachun, I. Manolescu, and M. Oulamara, *Rotational invariance in critical planar lattice models*, version 2, June 30, 2026. https://arxiv.org/abs/2012.11672v2.

[4] H. Duminil-Copin, K. K. Kozlowski, D. Krachun, I. Manolescu, and T. Tikhonovskaia, *On the six-vertex model's free energy*, version 2, August 24, 2022. https://arxiv.org/abs/2012.11675v2.

[5] E. Averous, H. Duminil-Copin, T. He, P. Lammers, and I. Manolescu, *Conformal invariance of the six-vertex model's height function*, October 5, 2026, version 1. https://arxiv.org/abs/2610.07454v1.

[6] M. Subbarao, *RamCalBooks*, Volume 21, original chapter `ch01d-physical-metrics.tex`, private local manuscript. Fresh literal scope: lines 1–210.

[7] M. Subbarao, *RamCalBooks*, Volume 21, original chapter `ch01f-spectral-limits.tex`, private local manuscript. Fresh literal scope: lines 1–242.

[8] M. Subbarao, *RamFoundationTRs*, original `TR-rev-1-old-tr117.tex`, private local manuscript. Fresh literal scope: lines 120–315 and 984–1126.